

Lesson 06.1: Bayesian Networks

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References:

- AIMA (Artificial Intelligence: a Modern Approach)
 - Chap 12, Quantifying uncertainty
 - Chap 13: Probabilistic reasoning
 - Chap 14: Probabilistic reasoning over time



- *Logic-Based AI Under Uncertainty*
- Probabilistic Reasoning

Logic-Based AI Under Uncertainty: Problem

- **Logic-based AI systems:**
 - Based on **propositional logic**
 - Represent **actions** using **rules** like:
 - * *"If preconditions P hold, then action A causes effect E "*
 - Example:
 - * *"If I turn the car key, the engine starts"*
 - * **But:** the battery might be dead, there's no fuel, the starter is broken, etc.
- Real-world agents face **uncertainty** from:
 - Partial observability
 - * Agent can't see the full state of the world
 - Non-determinism
 - * Actions don't always have predictable outcomes
 - Adversarial conditions
 - * Other agents may interfere

Logic-Based AI Under Uncertainty: Naive Solution

- A possible solution to uncertainty is:
 1. Use a **belief state**, i.e., set of all possible current world states
 2. Use **causal and exhaustive augmentation** for rules
 - Must consider all possible preconditions for acting, even unlikely ones
 3. Construct **contingent plans** that handle every possible sensor report and belief
 - Plans become large and complex
 - No guaranteed plan may exist, yet action is required

Logic-Based AI Under Uncertainty: Example

- **Goal:** start a car by turning the car key
- **Obvious preconditions**
 - The car has fuel
 - The car battery is charged
 - The car key is the right one
 - The starter motor is working
 - ...
- **Less obvious preconditions**
 - The fuel lines are not clogged
 - The fuel pump is working
 - The electrical system is intact (no blown fuses)
 - The engine oil level is sufficient
 - ...
- Every precondition requires something different to do to achieve the goal!

Causal and Exhaustive Augmentation

- To use propositional logic under uncertainty, augment the left-side of $X \implies Y$ to make it:
 1. **Causal**: identify true causal-effect relationships
 2. **Exhaustive**: identify all possible conditions leading to the outcome
- **Logical qualification problem**
 - Enumerate all the preconditions necessary for an action to succeed
- **Difficulties**
 - Every action can have more hidden conditions in order to succeed
 - * E.g., start the engine $\implies$ turn car key $\implies$ starter motor is working $\implies$ battery is charged $\implies$...
 - Preconditions can change depending on the situation
 - * E.g., start the engine $\implies$ battery is charged $\implies$ it's not too cold $\implies$...

Causal and Exhaustive Augmentation

- **Limitations** of logical qualification
 1. **Laziness**
 - Too much work to create all possible rules
 2. **Theoretical ignorance**
 - Science doesn't always have a complete theory of the domain
 - E.g., medical science doesn't know all the "rules"
 3. **Practical ignorance**
 - Even if you knew all the rules, you might not have all the information needed
 - E.g., not all necessary medical tests can be run for a particular patient
- Expert systems failure and AI winter (mid 1980s, 1990s)
 - The real world is complex and open-ended
 - Logical rules can't capture all necessary and sufficient conditions

Failure of Logic-Based AI: Wet Grass Example

- Consider the propositions:
 - *Rain* = “it rains”
 - *WetGrass* = “the grass is wet”
- You would expect $Rain \implies WetGrass$



- “ $Rain \implies WetGrass$ ” is not true in general
 - If it rains but there is a cover over the grass, the grass will not be wet
 - If it rains but there is high temperature, the wet grass might dry quickly
- “ $WetGrass \implies Rain$ ” is not true in general
 - The grass could be wet because of a sprinkler system
 - The grass could be wet because of morning dew
- You need to consider also the propositions:
 - *Cover* = “there is a protective cover over the grass”
 - *Evaporate* = “the water evaporates quickly”
 - *Sprinkler* = “the sprinkler system is on”
 - *Dew* = “there is morning dew”

Failure of Logic-Based AI: Wet Grass Example

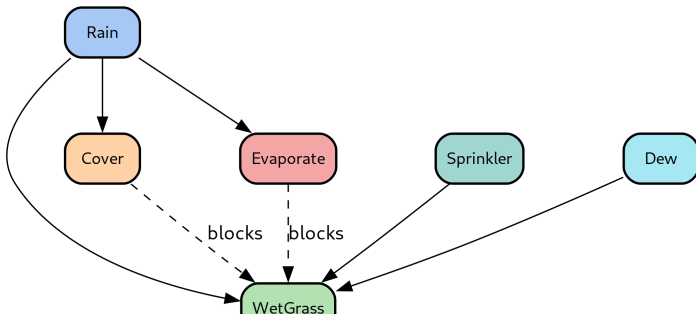
- Identify all exceptions and dependencies to make $X \implies Y$:

1. Causal

- “If it rains and there is no other source of water, the grass will be wet”
- $Rain \implies WetGrass \vee Cover \vee Evaporate \dots$
- $Rain \wedge \neg(Cover \wedge Evaporate \dots) \implies WetGrass$

2. Exhaustive

- “The grass is wet, if it rains or sprinkler is on or there is morning dew”
- $WetGrass \implies Rain \vee Sprinkler \vee Dew \dots$



Acting Under Uncertainty: Actual Solution

- Can't use propositional logic under uncertainty
- Acting under uncertainty requires combining:
 - **Probability**: to handle uncertainty and partial knowledge
 - **Utilities**: for evaluating desirability of each outcome
- **Key idea:**
 - Rational choice = plan that maximizes expected utility
 - * Performance measure: combines goals like punctuality, comfort, legal compliance
 - * Belief: agent's internal estimate of outcome likelihoods
 - Evaluate plans based on performance on average, given known information
 - But success is not guaranteed!

The Paradox of Probability and Knowledge

- **Paradox**
 - *“There is no uncertainty in the actual world!”*
 - E.g., the grass is wet, but either it has rained or not
- **Knowledge is subjective**
 - Probabilities relate to a knowledge state, not to the real world
 - Updating knowledge changes probability statements

The Paradox of Probability and Knowledge

- E.g., updating belief about wet grass and rain
- Initially, you observe wet grass
 - From past data you know that $\Pr(Rain|WetGrass) = 0.8$
 - 80% chance it rained if grass is wet
 - This is the Bayesian prior
- You learn new information:
 - Sprinkler was on
 - Wet grass could be due to the sprinkler, not rain
 - Belief changes: $\Pr(Rain|WetGrass \wedge Sprinkler) = 0.4$
 - This is a Bayesian update
- You further observe:
 - Weather report says there was no rain
 - Certain it did not rain, despite wet grass
 - Evidence overrides prior: $\Pr(Rain|WetGrass \wedge WeatherReport) = 0$
- All the statements were true even if contradictory (given the state of knowledge!)

- Logic-Based AI Under Uncertainty
- ***Probabilistic Reasoning***
 - Conditional Independence
 - Bayesian Networks

- Logic-Based AI Under Uncertainty
- Probabilistic Reasoning
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Full Joint Probability Distribution

- Consider a set of random variables $X_1, X_2, \dots, X_n$
- The **full joint probability distribution**

- Assigns a probability to every possible world:

$$\Pr(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

where a **possible world** is a particular assignment of values to all variables

- Can answer any probabilistic query about the domain

- **Cons**

- Size grows exponentially k^n with the number of variables n and number of values k
- Impractical for real-world problems with many variables
- Manually specifying each entry is tedious

- **Independence** (conditional and absolute) simplifies modeling

- In the real world, many variables are not fully dependent on all others
- Reduces the number of variables needed in the model
- Makes compact and structured representations possible

Independence of Random Variables: Definition

- Two random variables X and Y are **independent** iff:

$$\Pr(X, Y) = \Pr(X) \cdot \Pr(Y)$$

- Equivalently, knowing Y tells nothing about X

$$\Pr(X|Y) = \Pr(X)$$

- E.g.,
 - The events “coin flip” and “weather” are independent

$$\Pr(\text{Coin} = \text{Heads} | \text{Weather} = \text{Rainy}) = \Pr(\text{Coin} = \text{Heads})$$

- Independence** of random variables
 - Reduces the number of parameters needed to model a system, e.g.,

$$\Pr(X_1 | X_2, X_3) = \Pr(X_1)$$

- Allows factorization of joint distribution, e.g.,

$$\Pr(X_1, X_2, X_3) = \Pr(X_1) \cdot \Pr(X_2) \cdot \Pr(X_3)$$

Conditional Independence: Definition

- Two random variables X and Y are **conditionally independent** given a random variable Z iff knowing Z makes X and Y independent:

$$X \perp Y|Z \iff \Pr(X, Y|Z) = \Pr(X|Z) \cdot \Pr(Y|Z)$$

- Example**

- $X = \text{"it is raining today"}$
- $Y = \text{"a person is carrying an umbrella"}$
- $Z = \text{"the weather forecast"}$
- Without Z** : there is a relationship between X and Y (i.e., X and Y are not independent)
- Given Z** : rain X may not directly influence whether a person carries an umbrella Y
- Thus, X and Y are conditionally independent given Z

- Pros**

- Conditional independence is more common than absolute independence
- Simplify probabilistic models by factorizing the joint conditional distribution into product of individual conditional distributions

Conditional Independence: Example

- Two events can become independent once we know a third event

- Example**

- Fire* = “there is a fire”
- Toast* = “someone burned toast”
- Alarm* = “the fire alarm rings”
- Call* = “a friend calls to check on you”

- Dependencies**

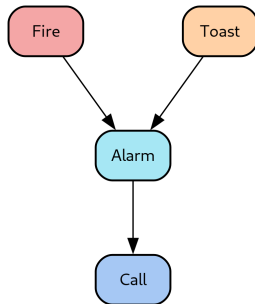
- Alarm* depends on *Fire* or *Toast*
- Call* depends on whether *Alarm* rings

- Conditional independence**

- Once we know the alarm rang, the specific cause doesn't affect whether the friend calls
- $\Pr(\text{Call} \mid \text{Alarm}, \text{Fire}) = \Pr(\text{Call} \mid \text{Alarm})$

- Interpretation**

- Call* is conditionally independent of *Fire* and *Toast* given *Alarm*
- Knowing the alarm rang “blocks” the path of influence from *Fire* and *Toast* to *Call*

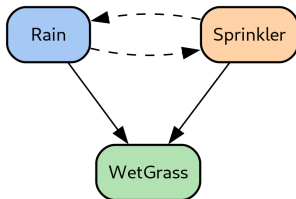


Conditional Independence: Garden Example

- Garden world with *Rain*, *Sprinkler*, and *WetGrass*
 - Is $\Pr(Rain|Sprinkler) = \Pr(Rain)$?
 - * **No**: if the sprinkler is on, it's less likely it rained
 - * *Rain* and *Sprinkler* are not independent
 - Is $\Pr(Rain|Sprinkler, WetGrass) = \Pr(Rain|WetGrass)$?
 - * **Yes**: knowing the grass is wet, whether the sprinkler was on tells you nothing more about the rain
 - * *Rain* and *Sprinkler* are conditionally independent given *WetGrass*

- **Interpretation:**

- Without *WetGrass*: *Rain* and *Sprinkler* affect each other because they both explain *WetGrass*
- With *WetGrass*: once *WetGrass* is observed, the “explaining away” effect occurs



- **"Explaining away" occurs when**

- Two variables (causes) independently influence a third variable (effect)

- Logic-Based AI Under Uncertainty
- Probabilistic Reasoning
 - Conditional Independence
 - *Bayesian Networks*

Bayesian Networks: Definition

- Aka:
 - “Bayes nets”,
 - “Belief networks”
 - “Probabilistic networks”
 - “Graphical models” (somehow a broader class of statistical models)
 - “Causal networks” (arrows have special meaning)
- A **Bayesian network** is a Directed Acyclic Graph (DAG)
 1. **Nodes** X_i correspond to random variables (discrete or continuous)
 2. **Edges** $X \rightarrow Y$ connect nodes and represent direct dependencies among variables
 - We say that $X = \text{Parent}(Y)$, $Y = \text{Child}(X)$, ancestors, spouses, ...
 3. Each node X_i is associated with a **conditional probability** (CPD):

$$\Pr(X_i | \text{Parents}(X_i))$$

quantifying the effect of the parents on the node

- If a node has no parents, it has a prior probability

Bayesian Network: Intuition

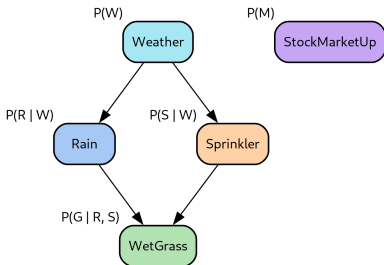
- Bayesian networks are the analogous for uncertain knowledge to propositional logic for definite knowledge
 - **Propositional logic** = rigid rules, i.e., **True** or **False**
 - **Bayesian networks** = flexible inference, i.e., degrees of belief
 - Replace $X \implies Y$ with $\Pr(Y|X)$
- E.g., wet grass example
 - R = “It is raining”
 - W = “The grass is wet”
 - **Propositional logic**
 - * If $R \rightarrow W$ and R is true, then W must be true
 - **Bayesian network**
 - * $\Pr(R = \text{True}) = 0.2$
 - * $\Pr(W|R) = 0.9$
 - * $\Pr(W|\neg R) = 0.1$
- E.g., medical diagnosis
 - **Propositional logic**
 - * “Patient has disease D ” $\implies$ “Patient has symptom S ”
 - **Bayesian network**
 - * “Probability of S given D is high, but not certain”

Bayesian Network and Full-joint distribution

- It can be shown that **topology** and **conditional probabilities** are sufficient to specify the full joint distribution
 - **Any full joint** distribution
 - **Very concisely** (often)
- Nodes are:
 - Directly influenced by their parents
 - Indirectly influenced by all their ancestors
- The topology of the network (nodes and edges) specifies conditional independence relationships
 - E.g., $X \rightarrow Y$ means “*X has a direct influence on Y*”, i.e., “*X relates to Y*” (not necessarily “causes”)
- How to build a Bayesian Network:
 - Domain experts can decide what relationships exist among domain variables, determining the topology
 - Conditional probabilities can be specified or estimated

Bayesian Networks: Wet Grass Example

- Consider a world with 5 variables
 - *Weather*
 - * Represents general environmental conditions (e.g., sunny, cloudy)
 - * Can't be observed
 - *Rain*
 - * Directly influenced by *Weather*
 - *Sprinkler*
 - * Influenced by *Weather* (e.g., less likely when raining)
 - *WetGrass*
 - * Represents whether the grass is wet
 - * Affected by both *Rain* and *Sprinkler*
 - *StockMarketUp*
 - * Indicates whether the



Bayesian Networks: Wet Grass Example

- *Rain* and *Sprinkler* are **dependent** (marginally)

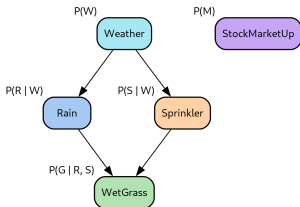
$$\text{Rain} \not\perp \text{Sprinkler} \iff \text{Rain} \leftrightarrow \text{Sprinkler}$$

- If there is no *Rain*, *Weather* = *sunny* and *Sprinkler* is more likely to happen (since sprinklers are usually turned on in sunny weather)
- *Rain* and *Sprinkler* are related since they share a common parent *Weather*

- *Rain* and *Sprinkler* are **conditionally independent** given *Weather*

$$\text{Rain} \perp \text{Sprinkler} | \text{Weather}$$

- Once *Weather* is fixed, knowing whether it rained tells you nothing about the *Sprinkler*
- The correlation is “explained away” by *Weather*



Bayesian Networks: Wet Grass Example

- *Rain* and *Sprinkler* are **conditionally independent** given *WetGrass*, but only if *Weather* is not observed

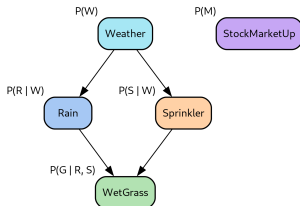
$$Rain \perp Sprinkler | WetGrass(notWeather)$$

- If we condition on *WetGrass*, i.e., the grass is wet, knowing that the sprinkler was off increases the chances that it must have rained (and vice versa)

- *Rain* and *Sprinkler* are **conditionally dependent** given *WetGrass*, and *Weather*

$$Rain \not\perp Sprinkler | WetGrass, Weather$$

- If you know that it's sunny, rain is unlikely and sprinkler likely
- If you also know that *WetGrass*
 - * If you learn *Sprinkler* off $\implies$ must have *Rain*
- Fork path (common cause) and collider path (common effect)
- *StockMarketUp* is **(unconditionally) independent** of all other variables



Conditional Probability Table

- **Conditional Probability Table** (CPT) encodes the probability of one node X_i given its parents $Parents(X_i)$

$$\Pr(X_i | Parents(X_i))$$

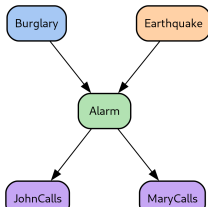
- Each row of the CPT contains the conditional probability of the node under a conditioning case (i.e., a possible combination of the values for the parent nodes)
- E.g., $\Pr(A|B, C)$

A	P(A B,C)	P(A B,-C)	P(A -B,C)	P(A -B,-C)
True	0.80	0.10	0.05	0.05
False	0.05	0.85	0.10	0.0

- **Note:**
 - Natural for discrete variables, but can be extended to continuous variables
 - A conditional probability table summarizes an infinite set of circumstances in the table

Bayesian Networks: Burglar Example

- Famous example from Judea Pearl
- An *Alarm* system installed at a home in LA
 - Fairly reliable at detecting *Burglary*
 - Also responds to minor *Earthquakes* (false positive)
- Two neighbors, *John* and *Mary* will *Call* you when they hear the *Alarm*
 - *John*:
 - * Almost always *Calls* when he hears the alarm
 - * Sometimes confuses telephone with the *Alarm* and *Calls* (false positives)
 - *Mary*:
 - * Misses the alarm 30% of the cases (false negatives)
- The **structure of the graph** shows that:
 - *Burglary* and *Earthquake* affects the event *Alarm*
 - *JohnCalls* and *MaryCalls* depend only on the *Alarm*, and not on *Burglary* and *Earthquake*



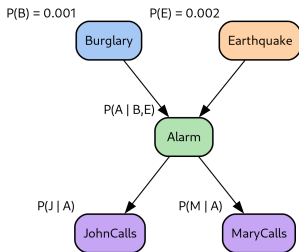
Bayesian Networks: Burglar Example

- The probability of *Burglary* is 0.001
- The probability of *Earthquake* is 0.002
- *Alarm* system
 - Is fairly reliable at detecting *Burglary*
 - Responds to minor *Earthquakes*

Burglary	Earthquake	P(Alarm B,E)
True	True	0.95
True	False	0.90
False	True	0.30
False	False	0.01

- Since events are independent:

$$\Pr(\text{Alarm}) = \Pr(\text{Alarm} | \text{Burglary}, \text{Earthquake}) \Pr(\text{Burglary}) \Pr(\text{Earthquake})$$



Bayesian Networks: Burglar Example

- Two neighbors, *John* and *Mary* will *Call* you when they hear the *Alarm*
 - John*:
 - * Almost always *Calls* when he hears the alarm
 - * Sometimes confuses telephone with the *Alarm* and *Calls* (false positives)
 - Mary*:
 - * Misses the alarm 30% of the cases (false negatives)
- JohnCalls* and *MaryCalls* are represented by:

Alarm (A)	P(<i>JohnCalls</i> Alarm)
True	0.90
False	0.05

Alarm (A)	P(<i>MaryCalls</i> Alarm)
True	0.70
False	0.01

Conditional Probability Table

- A **node without parents** has an unconditional probability

P(Burglary)
.001

- The **sum of probabilities** must be 1
 - If there is a single input variable, it is possible to remove the redundancy

Alarm (A)	P(JohnCalls .)	P(- JohnCalls .)
True	0.90	0.10
False	0.05	0.95

Alarm (A)	P(JohnCalls .)
True	0.90
False	0.05

- A **node with k parents** has 2^k possible rows in the table

Burglary	Earthquake	P(Alarm .)
T	T	.95
T	F	.94
...	...	...